

FRONT END DEVELOPMENT FRAMEWORKS AND UI ENGINEERING (25CS1201E)
2025-26, Term-III
Project Problem Statement Submission Form

Sec No-	Team No-	Project Title-<u>Hotel-Room-Booking-Engine</u>
Team Lead Roll Number	G teja	
Team Lead Name	2520090162	
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1. Abstract

The Hotel Room Booking Engine is a web-based application designed to simplify and streamline the process of searching and booking hotel rooms. This system allows users to explore available rooms based on selected dates, occupancy, and preferred amenities, while providing real-time availability and dynamic pricing. The platform ensures a user-friendly experience through features such as an interactive date picker, advanced filtering options, detailed room views, and a seamless booking process.

The application is built using modern web technologies including HTML5, CSS3, JavaScript (ES6+), React.js for the frontend, and Node.js with Express.js for backend services. RESTful APIs are used to manage communication between the client and server, while tools like Axios and Postman support API integration and testing. Optional database support using MySQL enables persistent storage of booking data, user history, and administrative controls.

Key modules of the system include room search, booking management, payment simulation, confirmation generation, and an admin panel for managing listings and reservations. The system also leverages LocalStorage or state management tools such as Context API or Redux to maintain user session data. Overall, the Hotel Room Booking Engine provides a scalable and efficient solution for online hotel reservations, improving user convenience and offering a foundation for real-world travel and hospitality applications.

2. Module Mapping

(Provide details of how modules are divided among team members.)

Module Name	Assigned Roll Number(s)	Signature
☐ Module 2 — Dashboard / Home Screen ☐ Module 3 — Core Feature A (Search & Booking UI) ☐ Module 5 — Form Handling ☐ Module 6 — Data Display	2520030539	
☐ Module 1 — Authentication & Authorization	2520090162	

☐ Module 4 — Core Feature B (Tracking & Management) ☐ Module 10 — Admin Panel (Backend logic)		
☐ Module 7 — Notifications & Alerts ☐ Module 8 — File Handling ☐ Module 9 — API Integration (Axios + Postman testing)	2520030468	

3. Module Details

Module 1 — Authentication & Authorization:

This module ensures secure access to the system by enabling user login and admin authentication. It manages user sessions using JWT or session-based methods and provides role-based access control for passengers and administrators.

Module 2 — Dashboard / Home Screen:

This module provides the main landing page of the system. It displays an overview of the feedback system and offers quick navigation options for submitting and viewing feedback.

Module 3 — Core Feature A (Feedback Submission):

This module allows passengers to submit feedback by providing ratings for various services such as cabin crew, food, cleanliness, and punctuality, along with additional comments.

Module 4 — Core Feature B (Feedback Analysis):

This module processes and analyzes feedback data. It includes sentiment classification and generates performance insights, which can be visualized using charts and graphs.

Module 5 — Form Handling:

This module manages feedback forms with proper validation. It ensures that all required fields are correctly filled and provides error messages and success confirmations.

Module 6 — Data Display:

This module is responsible for presenting feedback data in an organized manner using tables, lists, or card formats. It may include features like sorting, filtering, and pagination.

Module 7 — Notifications & Alerts:

This module provides user feedback through notifications such as confirmation messages (e.g., “Feedback submitted successfully”) and alert messages for errors or system updates.

Module 8 — File Handling:

This module supports exporting feedback data in formats such as CSV for reporting and record-keeping purposes.

Module 9 — API Integration:

This module connects the frontend (React) with the backend (Node.js/Express) using REST APIs. It handles data exchange, including sending requests and receiving responses with proper error and loading states.

Module 10 — Admin Panel:

This module provides administrative functionality, allowing admins to view, manage, and analyze all feedback entries, ensuring effective system control and monitoring.

4. Frameworks / Platforms / Software Used



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Frontend Technologies:

- **HTML5, CSS3, JavaScript (ES6+)** – For structuring, styling, and implementing interactive features of the web application
- **React.js** – For building dynamic and responsive user interfaces using component-based architecture
- **Bootstrap** or **Tailwind CSS** – For responsive design and pre-built UI components

Backend Technologies:

- **Node.js** – For server-side execution of JavaScript
- **Express.js** – For building RESTful APIs and handling server-side logic

Database:

- **MySQL** – For storing user data, feedback, and system records

API & Communication Tools:

- **Axios** – For making API requests from frontend to backend
- **Postman** – For testing and validating API endpoints

Data Visualization:

- **Chart.js** or **Recharts** – For displaying analytics and feedback insights through charts

Version Control & Development Tools:

- **Git** – For tracking changes in the project
- **GitHub** – For collaboration and code management
- **Visual Studio Code** – For writing and managing source code

Platform:

- Web-based application accessible through modern browsers like Chrome, Edge, or Firefox

Guide Name and Signature (remarks if any) _____